📊 Data Science Workflow Canvas*

Start here. The sections below are ordered intentionally to make you state your goals first, followed by steps to achieve those goals. You're allowed to switch orders of these steps!

Title: Climate Adaptation Finance in Thailand

1 Problem Statement

What problem are you trying to solve?
What larger issues do the problem address?

Questions

- How is climate adaptation funding distributed across Thailand?
- Does the funding reach the areas and sectors most vulnerable to climate change?

Larger Issues

- Funding may be concentrated in major cities.
- High investment does not always mean effective adaptation.

2 Outcomes/Predictions

What prediction(s) are you trying to make?
Identify applicable predictor (X) and/or target (y) variables.

Primary Outcomes

- Identify the total amount of climate adaptation funding.
- Compare funding across provinces and sectors.

Predictions

- Water management will receive the largest share of funding.
- Some provinces may have many projects but relatively low funding.

Type

- Descriptive data analysis
- Exploratory data analysis

3 Data Acquisition

Where are you sourcing your data from?
Is there enough data? Can you work with it?

Data Sources

- Thailand Climate Finance Tracker 2026
- Data collected by CFNT and PIER
- **Coverage** : 2018–May 2026
- **Variables** : province, sector, funding amount, project count, funding source, and year

4 Modeling Data Analysis

What models are appropriate to use given your outcomes?

- Use **descriptive and exploratory data analysis.**
- Apply **grouping and aggregation** to summarize funding by province and sector.
- Use **percentage analysis** to compare funding proportions.
- Apply **ranking analysis** to identify the highest- and lowest-funded provinces.
- Use **cross-tabulation** to compare provinces, sectors, and project numbers.
- Conduct **regional comparison** to identify funding patterns and inequalities.
- No predictive machine-learning model is used.

5 Model Evaluation Data Visualization

How can you evaluate your model's performance?

- **Bar charts**: funding by province and sector
- **Choropleth map**: funding distribution across Thailand
- **Treemap**: funding share by adaptation sector
- **Scatter plot**: number of projects versus funding amount
- **Stacked bar chart**: public and private investment



6 Data Preparation

What do you need to do to your data in order to run your model and achieve your outcomes?

Steps:

- Remove duplicate records.
- Handle missing or undisclosed values.
- Standardize province and sector names.
- Convert all funding amounts into the same unit.
- Group data by province, sector, and year.
- Calculate totals, percentages, and rankings.
- Check for inconsistent project classifications.
- Verify that the final dataset matches the original source.

✅ Activation

When you finish filling out the canvas above, now you can begin implementing your data science workflow in roughly this order.

1 Problem Statement → 2 Data Acquisition → 3 Data Prep → 4 Modeling → 5 Outcomes/Preds → 6 Model Eval

Conceptualized by Jasmine Vasandani using notes from General Assembly's Data Science Immersive. Format inspired by Business Model Canvas.

6730084521 Chatrphol Ovanonchai

2107540 Data Science for Environmental Apps

Final Project Part B